

No. of Functional Features Included in the Solution

Date	03 JULY 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	HDI Input Form	Accepts Life Expectancy, Mean Schooling, Expected Schooling, and GNI per Capita.	Prediction UI	Done	High
2	Input Validation	Checks user-entered values before prediction.	Flask Backend	Done	Medium
3	Data Preprocessing	Prepares and transforms input data for the trained model.	preprocessing.py	Done	High
4	Feature Scaling	Applies the saved scaler to input features before prediction.	scaler.pkl	Done	High
5	HDI Score Prediction	Predicts the Human Development Index score using the trained model.	model.pkl / Prediction Engine	Done	High
6	HDI Category Classification	Classifies the result into Very High, High, Medium, or Low human development.	prediction.py	Done	High
7	Prediction Result Display	Displays the predicted HDI score and category to the user.	result.html	Done	Medium
8	Data Visualization	Displays EDA plots and development indicator relationships.	Plots / Home Page	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8

Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	HDI Input Form, Prediction Result Display
2	Backend / Logic	5	Input Validation, Preprocessing, Feature Scaling, HDI Prediction, Category Classification
3	Database / Storage	1	Local storage of dataset, trained model, scaler, metrics, and plots
4	API / Integration	0	Not Applicable – no external API integration is required
5	Security / Authentication	0	Not Applicable – the application does not collect user accounts or sensitive personal data